Lecture 36

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1 Convergence and Boundedness

1.1 Question

Is every bounded sequence convergent?

- **Answer**: No.

- **Counter-example**: Let $a_n = (-1)^n$. The sequence is bounded ($-1 \leq a_n \leq 1$) but does not converge.

2 Arithmetic of Limits

2.1 Theorem

Let (a_n) and (b_n) be convergent sequences such that $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$. Then:

1. $\lim_{n\to\infty}(a_n + b_n) = a + b$
2. $\lim_{n\to\infty}(c \cdot a_n) = c \cdot a$ for any $c \in \mathbb{R}$
3. $\lim_{n\to\infty}(a_n \cdot b_n) = a \cdot b$
4. If $b \neq 0$ and $b_n \neq 0$, then $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}$

2.2 Proof of Property (3): Product Rule

We want to show that for any $\epsilon > 0$, there exists an N such that $|a_n b_n - ab| < \epsilon$ for all $n \geq N$.

Using the triangle inequality:

$$|a_n b_n - ab| = |a_n b_n - ab_n + ab_n - ab|$$

$$\leq |b_n| \cdot |a_n - a| + |a| \cdot |b_n - b|$$

Since (b_n) is convergent, it is bounded. Let $M_b = \sup\{|b_n|\}$. Let $M = \max\{M_b, |a|\}$. Then:

$$|a_n b_n - ab| \leq M(|a_n - a| + |b_n - b|)$$

Since $a_n \to a$ and $b_n \to b$:

- $\exists N_1$ such that $\forall n \geq N_1, |a_n - a| < \frac{\epsilon}{2M}$.
- $\exists N_2$ such that $\forall n \geq N_2, |b_n - b| < \frac{\epsilon}{2M}$.

Let $N = \max\{N_1, N_2\}$. For all $n \geq N$:

$$|a_n b_n - ab| \leq M\left(\frac{\epsilon}{2M} + \frac{\epsilon}{2M}\right) = \epsilon$$

3 Examples and Common Pitfalls

3.1 Exercise

Calculate the limit:

$$\lim_{n\to\infty} \frac{2n^2+3}{n^2+n+1}$$

Divide numerator and denominator by n^2:

$$= \lim_{n\to\infty} \frac{2+\frac{3}{n^2}}{1+\frac{1}{n}+\frac{1}{n^2}} = \frac{2+0}{1+0+0} = 2$$

3.2 A Common Misconception

Consider:

$$1 = \lim_{n\to\infty} 1 = \lim_{n\to\infty} \underbrace{\left(\frac{1}{n}+\frac{1}{n}+\cdots+\frac{1}{n}\right)}_{n \text{ times}}$$

One might incorrectly apply the sum rule:

$$= \lim_{n\to\infty} \frac{1}{n} + \cdots + \lim_{n\to\infty} \frac{1}{n} = 0 + \cdots + 0 = 0$$

Result: $1 = 0$ (Contradiction). **Reason**: The arithmetic rule for the sum applies to a **fixed** number of sequences, not a sum where the number of terms grows with n.

4 The Squeeze Theorem (Sandwich Theorem)

4.1 Theorem

Let $(a_n), (b_n), (c_n)$ be sequences such that $a_n \leq b_n \leq c_n$ for all n. If $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} c_n = a$, then $\lim_{n\to\infty} b_n = a$.

4.2 Proof

We need to show $|b_n - a| < \epsilon$ for sufficiently large n. Given convergence of a_n and c_n:

- $\exists N_1$ such that $\forall n \geq N_1, |a_n - a| < \epsilon \iff -\epsilon < a_n - a < \epsilon$.
- $\exists N_2$ such that $\forall n \geq N_2, |c_n - a| < \epsilon \iff -\epsilon < c_n - a < \epsilon$.

Let $N = \max\{N_1, N_2\}$. For $n \geq N$:

$$-\epsilon < a_n - a \leq b_n - a \leq c_n - a < \epsilon$$

$$\implies -\epsilon < b_n - a < \epsilon$$

$$\implies |b_n - a| < \epsilon$$

5 Special Limits involving Roots

5.1 Theorem 1: n-th root of a constant

Let $a > 0$. Then $\lim_{n\to\infty} \sqrt[n]{a} = 1$.

5.1.1 Proof (for $a > 1$)

Let $\sqrt[n]{a} = 1 + x_n$ where $x_n > 0$. Raise to the power n:

$$a = (1 + x_n)^n$$

Using Bernoulli's Inequality $(1 + x)^n \geq 1 + nx$:

$$a \geq 1 + nx_n$$

$$nx_n \leq a - 1 \implies 0 < x_n \leq \frac{a-1}{n}$$

As $n \to \infty$, $\frac{a-1}{n} \to 0$. By the Squeeze Theorem, $x_n \to 0$. Thus, $\lim_{n\to\infty} \sqrt[n]{a} = \lim_{n\to\infty}(1 + x_n) = 1$.

5.2 Example

Calculate $\lim_{n\to\infty} \sqrt[n]{2^n + 7^n}$. Bounds:

$$\sqrt[n]{7^n} \leq \sqrt[n]{2^n + 7^n} \leq \sqrt[n]{7^n + 7^n}$$

$$7 \leq \sqrt[n]{2^n + 7^n} \leq 7 \cdot \sqrt[n]{2}$$

Since $\sqrt[n]{2} \to 1$, both bounds converge to 7.

$$\lim_{n\to\infty} \sqrt[n]{2^n + 7^n} = 7$$

5.3 Theorem 2: n-th root of n

$$\lim_{n\to\infty} \sqrt[n]{n} = 1$$

5.3.1 Proof

Let $\sqrt[n]{n} = 1 + x_n$ where $x_n \geq 0$.

$$n = (1 + x_n)^n$$

Using the Binomial Theorem, for $n \geq 2$:

$$n = 1 + nx_n + \binom{n}{2} x_n^2 + \cdots + x_n^n$$

Discarding positive terms:

$$n > \binom{n}{2} x_n^2 = \frac{n(n-1)}{2} x_n^2$$

$$1 > \frac{n-1}{2} x_n^2$$

$$x_n^2 < \frac{2}{n-1}$$

$$0 \leq x_n < \sqrt{\frac{2}{n-1}}$$

As $n \to \infty$, $\sqrt{\frac{2}{n-1}} \to 0$. Thus $x_n \to 0$. Therefore, $\lim_{n\to\infty} \sqrt[n]{n} = 1$.